```
    public Optional<Token> read(Request request, String
tokenId) {

        var session = request.session(false);
        if (session == null) {
            return Optional.empty();
        }

        var provided = Base64url.decode(tokenId);    #B
        var computed = sha256(session.id());    #B

        if (!MessageDigest.isEqual(computed, provided)) {
#C
            return Optional.empty();    #C
        }

        var token = new Token(session.attribute("expiry"),
                session.attribute("username"));
        token.attributes.putAll(session.attribute("attrs"));

        return Optional.of(token);
    }

    static byte[] sha256(String tokenId) {
        try {
            var sha256 = MessageDigest.getInstance("SHA-
256");    #D
            return sha256.digest(    #D
                tokenId.getBytes(StandardCharsets.UTF_8));
#D
        } catch (NoSuchAlgorithmException e) {
            throw new IllegalStateException(e);
        }
    }
}
```